CubeMate

Bill of Materials and Mechanical Assembly Guide

Last Revised 9th November 2027

Scope: This document covers the mechanical assembly of CubeMate, installation of the already-built controller PCB onto the top plate, and the published payload / EQ-wedge interface plates. Construction and wiring details for the custom controller PCB are intentionally outside this guide and will be documented separately.



Figure 1 - Cubemate Main Plates

Bill of Materials

Quantities below are for one complete two-axis CubeMate using the demonstrated 80T : 16T, 5:1 belt reduction on both RA and DEC. Alternative main-pulley sizes are listed separately and are substitutions, not additional parts.

Category	Item	Qty	Specification / notes
Machined structure	RA base / RA drive plate	1	Primary bottom plate; machined harmonic-drive seat and bearing location; no rebated edge lip.
Machined structure	DEC drive plate	1	Axis plate; mechanically the same drive assembly as RA, but with a rebated locating lip for positive engagement with the RA base plate.
Machined structure	Rear / counterweight plate	1	Outer plate with counterweight mounting point and rebated locating lip.
Machined structure	Left side plate	1	Machined side plate with rebated locating geometry; current STEP file engraved LEFT.
Machined structure	Right side plate	1	Machined side plate with rebated locating geometry; current STEP file engraved RIGHT.
Machined structure	Top plate	1	Rebated top closure plate, machined for the controller PCB, power connector and USB-C connector.
Machined structure	Motor mounting plate	2	Identical RA and DEC internal motor plates; each accepts one 5 mm bearing and five standoffs.
Drive train	Type 14 cup-style harmonic drive [for 42 Stepper – must use 5mm shaft]	2	https://www.aliexpress.com/item/1005006190112225.html
Drive train	5 mm D-Cut Drive Shaft	2	50–60 mm long, cut to suit. 60 mm works mechanically but protrudes farther than necessary. Smooth shaft can be used but is prone to slippage.
Drive train	Nema 17 Bipolar 0.9deg 36Ncm(51oz.in) 0.9A 42x42x40mm 4 Wires part code 17HM15-0904S	2	https://www.stepperonline.co.uk/nema-17-bipolar-0-9deg-36ncm-51oz-in-0-9a-42x42x40mm-4-wires-17hm15-0904s.html
Drive train	GT2 16-tooth motor pinion	2	5 mm bore.
Drive train	GT2 80-tooth main pulley / gear	2	5 mm bore. Demonstrated configuration; gives 5:1 reduction with 16T motor pinion.
Drive train	GT2 60-tooth main pulley / gear GT2 48-tooth main pulley / gear	Optional substitute	5 mm bore; 60T gives 3.75:1 reduction with 16T pinion. 48T gives 3:1. Enables easier belt-tension access than 80T. The lower reduction slightly reduces theoretical drive resolution.
Drive train	GT2 drive belt, 200 mm	2	Specified for the demonstrated 80T/16T arrangement.
Bearings / spacers	Flanged bearing	4	5 mm ID × 14 mm OD × 5 mm depth, with 1 mm locating flange. Two per axis. https://www.amazon.co.uk/dp/B0CRLGHJG9
Bearings / spacers	Nylon pulley spacer washer	10	5 mm ID × 1 mm thick. Five per axis; distribute above/below the main pulley as required for pulley alignment.

Category	Item	Qty	Specification / notes
Standoffs	Motor-plate standoff	10	M3 female–female × 24 mm. Steel preferred over brass for rigidity; five per axis.
Standoffs	Controller-PCB standoff	4	M3 female–female, 10–24 mm acceptable; 24 mm used in the demonstrated build.
Fasteners	M4 × 20 mm button-head machine screw	12	Six per harmonic drive, through RA/DEC plate into the harmonic-drive threaded holes.
Fasteners	M3 × 10 mm button-head machine screw	10	Five per axis; secures motor-plate standoffs to the motor mounting plate.
Fasteners	M3 × 16 mm countersunk machine screw	10	Five per axis; secures each motor mounting plate/standoff assembly to its RA or DEC plate.
Fasteners	M3 × 12 mm button-head machine screw	8	Four per motor; screws through the motor plate into the NEMA 17 motor.
Fasteners	M3 nut used as rigid spacer	8	Four per motor. Used loose as a spacer, not as a conventional nut; optional if the standoff mounting holes are countersunk.
Fasteners	M3 washer used with motor spacer	8	Four per motor; paired with spacer nuts in the demonstrated build.
Fasteners	M4 × 16 mm countersunk machine screw	18	3 DEC-to-RA + 3 rear-to-RA + 6 telescope dovetail + 6 chosen RA-to-wedge interface plate.
Fasteners	M4 × 12 mm countersunk machine screw	24	18 for the two side plates (9 each) + 6 for the top plate.
Fasteners	M3 × 8 mm countersunk machine screw	4	Secures controller-PCB standoffs to the top plate.
Fasteners	M3 × 8 mm button-head machine screw	4	Secures the controller PCB to its four standoffs.
Electronics / interfaces	Custom CubeMate controller PCB	1	Preassembled controller PCB with ESP32-S3; electronics construction documented separately.
Electronics / interfaces	5.5mm x 2.1mm Panel-mount DC power connector	1	https://www.amazon.co.uk/Lilyphon-Connectors-Systems-Electronics-Installation/dp/B0GY86BJWK
Electronics / interfaces	Panel-mount USB-C connector (4 wire only, can be micro or mini)	1	https://www.amazon.co.uk/dp/B0CPLRH4W6
Electronics / interfaces	Removable wiring headers / connectors	As required	Builder's choice. Demonstrated build uses IDC pin headers and JST-XH plug-in headers so panel connectors can be disconnected.
Payload interface	Vixen-style telescope dovetail mount	1	STEP file supplied; attaches to DEC axis with six M4 × 16 countersunk screws.
EQ-wedge interface	Vixen dovetail plate OR Losmandy plate	1	Choose one published STEP interface for the RA axis, or use a custom application-specific plate. Six M4 × 16 countersunk screws.
EQ-wedge interface	EQ wedge	1	User-selected. The prototype custom CubeMate wedge is intentionally not part of the published design at present.

Tools and consumables

- Hex keys / drivers for the selected M3 and M4 socket-head fasteners. The demonstrated build uses hex-drive hardware throughout.
- G-clamp, vice or other controlled press for bearing insertion. A tube and hammer can be used, but carries a greater risk of bearing damage.
- Soft protective layer between tools/clamps and anodised parts to avoid marking the finish.
- Engineer's square or equivalent squaring tool for final frame alignment.
- Small flat spanner (useful if the M3 spacer nuts bind while tightening the motor).
- Grease supplied with the harmonic drive, or a suitable alternative.
- Thread-locking compound is optional; it has not proved necessary in the demonstrated build.
- Optional drill/countersink tools if modifying the motor plates or drilling the threads out of spacer nuts.

Assembly Instructions

General fastening note: Unless a step explicitly says otherwise, leave the outer structural plate fasteners only snug during initial fit-up. Final tightening is performed in a controlled squaring sequence. Tighten firmly, but do not over-torque the aluminium threads or countersunk joints.

Finish protection: Use a soft protective layer wherever clamps, vices or other tools contact the anodised aluminium. The prototype used tissue paper; any clean, non-abrasive protective layer is suitable.

1. Assemble the RA drive module#



1.1 Press in the RA-base and motor-plate bearings

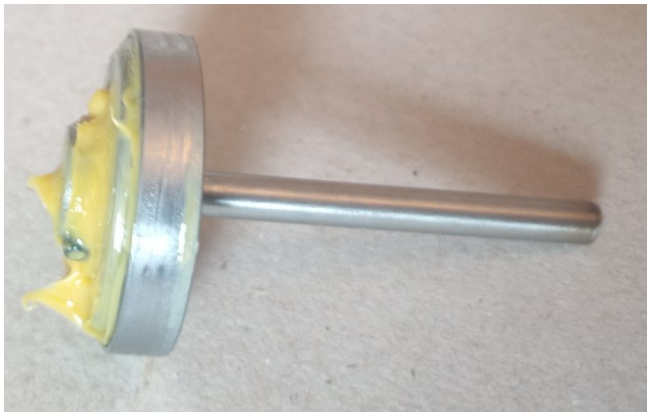
1. Press the first 5 × 14 × 5 mm flanged bearing into the central bearing seat in the RA base plate. The machining determines the correct insertion direction: press it fully into the recess so the locating flange sits flush with the machined surface.
2. Press the second bearing into the central hole of the motor mounting plate. Its orientation is not functionally important; the flange will rest against one face and the bearing will project approximately 1 mm through the opposite face.
3. Use a G-clamp, vice or other controlled pressing method where possible. A hammer and suitable tube can work, but is more likely to damage the bearing.





1.2 Fit the harmonic-drive wave generator and cup

4. Secure the harmonic-drive wave generator to the 5 mm drive shaft using the wave generator's own grub/set screws.
5. Ensure the inside of the harmonic-drive cup is well greased. Use the grease supplied with the drive, or a suitable alternative. The drive design permits later greasing from outside, but doing it now is easier.
6. Place the cup-style Type 14 harmonic drive into its machined seat in the RA base plate and feed the 5 mm shaft through the bearing.
7. From the shaft side of the RA plate, install six M4 × 20 mm button-head machine screws into the threaded harmonic-drive holes. Tighten evenly to clamp the drive into its seat. Button heads are preferred for clearance.







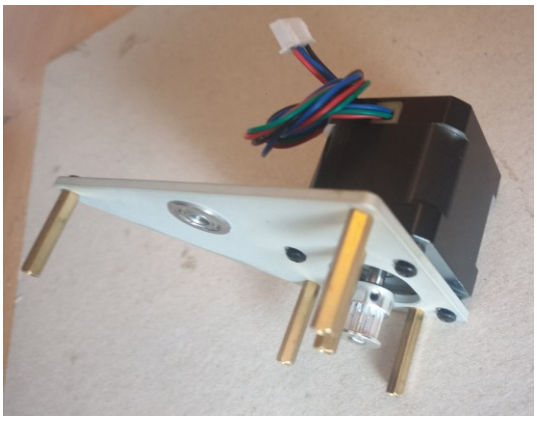
1.3 Fit the motor-plate standoffs

8. Orient the motor mounting plate as shown in the build photographs. The plate only fits the completed RA assembly in one geometry, but the early plates are not direction-marked.
9. Attach five M3 female–female × 24 mm standoffs to the motor mounting plate using five M3 × 10 mm button-head screws. All five standoffs project from the same face shown in the photographs.
10. Tighten the standoff screws firmly. Thread-locking compound may be used, but is optional.



1.4 Mount the NEMA 17 motor

11. The NEMA 17 mounts on the face of the motor plate opposite the five standoffs.
12. For the demonstrated non-countersunk motor plate, insert four M3 × 12 mm button-head screws through the slotted motor holes. On the motor side of the plate, place one M3 washer and one M3 nut over each screw as a rigid spacer.
13. The M3 nuts are spacers only: do not tighten them as conventional nuts. The screws continue through the spacers and tighten directly into the threaded NEMA 17 motor face.
14. Do not fully tighten the motor yet. Leave enough freedom for the motor to slide laterally in the slotted holes so that the GT2 belt can be tensioned.
15. If a spacer nut binds while tightening, a thin flat spanner can be used to free it. An optional modification is to drill the M3 thread out of the spacer nuts so they behave as plain bushes.
16. Alternative: countersink the five standoff mounting holes in the motor plate so their screw heads sit flush. If this is done, the motor can mount directly to the plate and the M3 nut/washer spacers are unnecessary.



1.5 Initial main-pulley positioning

17. Place a 5 mm ID × 1 mm nylon spacer washer on the RA shaft above the RA-plate bearing as an initial spacing reference.
18. Slide the 80T main pulley onto the 5 mm shaft with its hub/grub-screw side closest to the RA plate and the broad gear face farther away from the RA plate.
19. Do not tighten the 80T pulley grub screws yet. Pulley alignment is established first.
20. The final spacer distribution depends on the axial position of the 16T motor pinion. The aim is simply to align the 16T and 80T pulley tooth planes.

1.6 Test-fit the motor plate and set belt tension

21. Loop the 200 mm GT2 belt around the 16T motor pinion and the 80T main pulley.
22. Slide the motor-plate assembly down over the 5 mm shaft so that its bearing engages the shaft, then align the five standoffs with their mounting holes in the RA plate.
23. Temporarily secure at least three standoffs with M3 × 16 mm countersunk screws. Use three points spread around the pattern to triangulate the assembly. Tighten only snugly: the purpose is to hold the motor plate rigidly while belt tension is set.
24. Slide the motor across its slotted mounting holes until the GT2 belt has sensible working tension. Do not over-tension the belt; it only needs positive tooth engagement without excessive preload.
25. Tighten the motor mounting screws that are accessible. With the demonstrated 80T main pulley, only three of the four screws can be reached while assembled. With a 60T or 48T pulley, all four can be tightened in situ and the removal/reassembly steps below are unnecessary.

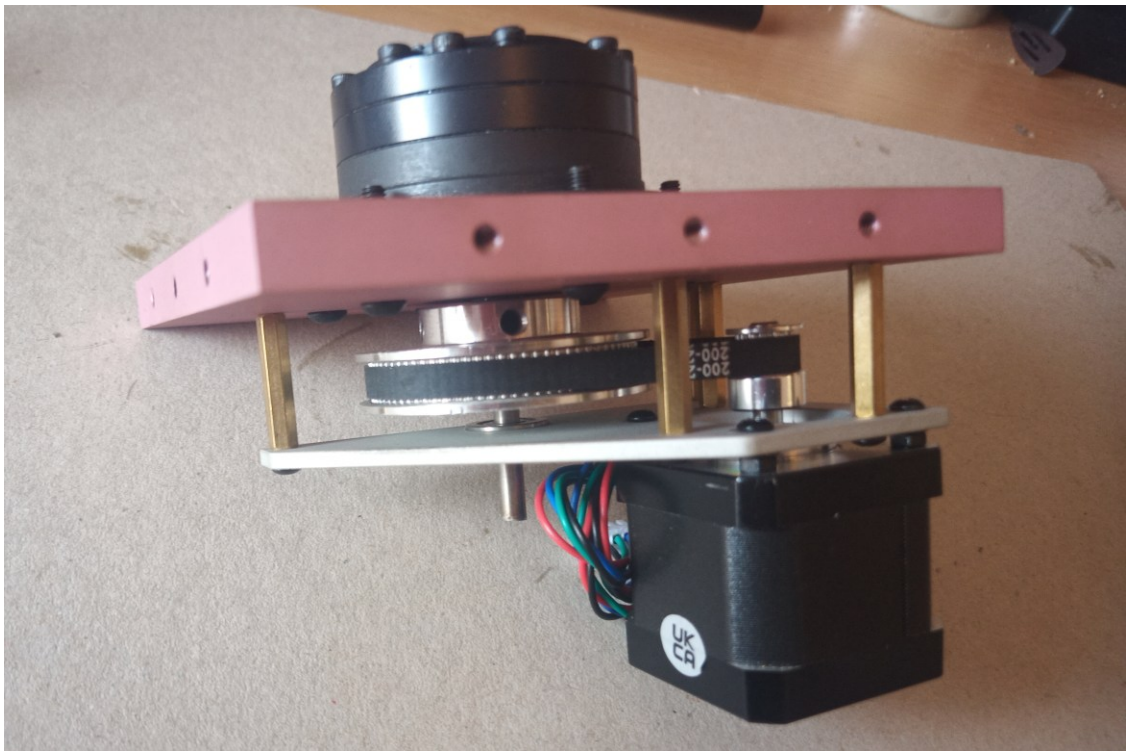
1.7 80T-only: remove the motor assembly and lock the motor position

26. Remove the three temporary M3 × 16 mm motor-plate fixing screws.
27. Slide the motor-plate assembly back off the 5 mm shaft. The tensioned belt prevents clean removal while the 80T pulley remains on the shaft, so allow the 80T pulley to slide off the shaft with the belt still looped around it.
28. Once clear of the RA shaft, remove the 80T pulley and belt from the motor assembly as needed for access.
29. Tighten the previously inaccessible fourth motor mounting screw, then snug all four motor screws firmly. The motor position now preserves the belt tension established during the test fit.

1.8 Final 80T pulley and belt installation

Spacer stack: Use five 1 mm nylon spacer washers per axis in total. Their split above and below the 80T pulley depends on the axial position of the 16T motor pinion. The prototype uses one below and four above; another build might use two below and three above. The objective is pulley alignment and prevention of unwanted axial movement.

30. Place the required lower nylon spacer washer(s) on the installed 5 mm RA shaft, then slide the 80T pulley onto the shaft. Keep the pulley grub screws loose.
31. Fit the remaining nylon spacer washers above the 80T pulley so the total spacer count for the axis is five.
32. Preferred 80T method: loop the belt around the 16T motor pinion first. Bring the motor assembly into position, place the belt over the RA shaft and engage the belt over part of the 80T pulley circumference.
33. Rotate the 80T pulley to “walk” the belt over the remaining teeth. The belt has just enough compliance to snap into full engagement without dismantling the motor mounting.
34. Move the motor-plate assembly into line with the RA shaft, ensure the upper bearing is accurately aligned, then slide the complete assembly down onto the shaft and standoffs.
35. Alternative method: preassemble the pulley/belt/spacer relationship on a spare piece of 5 mm shaft, align it precisely with the installed shaft, and slide the assembly into place. Take care to keep both shafts perfectly aligned.
36. Install all five M3 × 16 mm countersunk motor-plate screws and tighten them fully.
37. Verify that the two pulleys are axially aligned, the GT2 belt is correctly seated, and the belt tension remains appropriate. Only then tighten the 80T pulley’s own grub/set screws onto the 5 mm drive shaft.



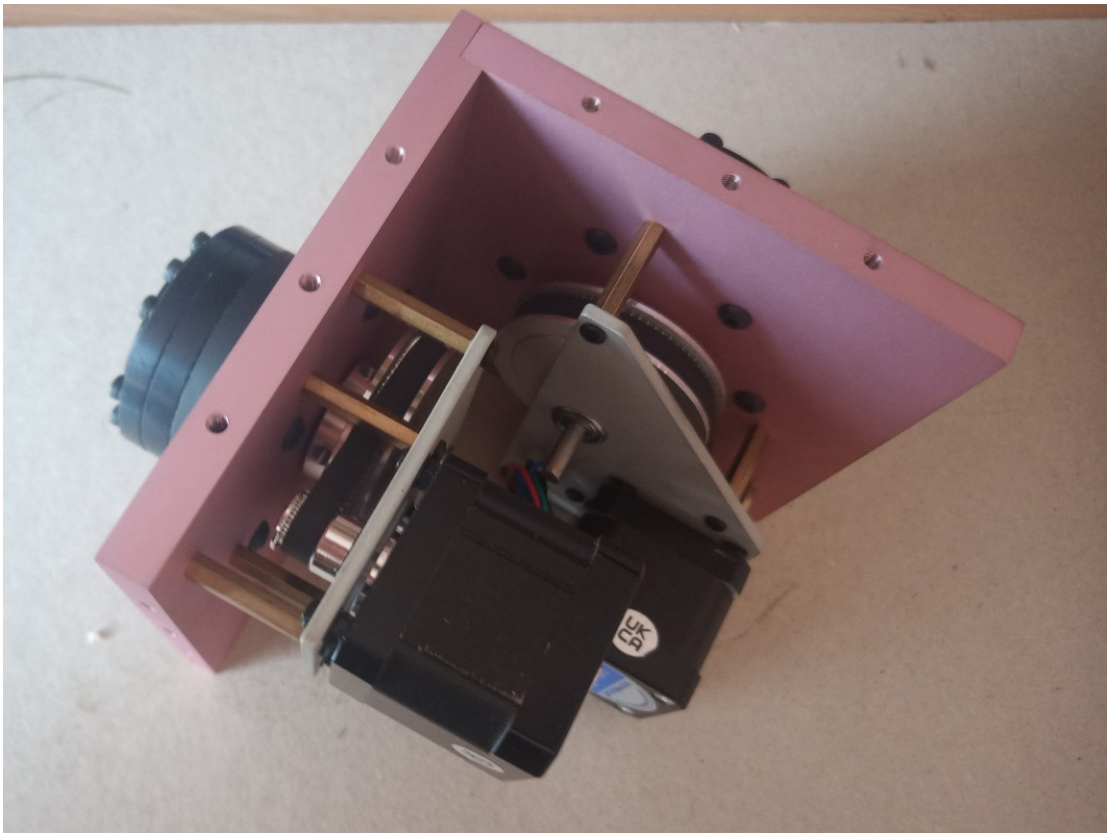
2. Assemble the DEC drive module

Repeat Section 1 in full for the DEC plate. The DEC drive assembly is a complete duplicate of the RA drive assembly: same Type 14 harmonic drive, shaft, bearings, motor mounting plate, NEMA 17 motor, pulleys, belt, standoffs, spacers and fasteners. **The only functional plate difference is the DEC plate’s rebated edge lip, which positively locates it onto the RA base plate during cube assembly.**

3. Assemble and square the CubeMate outer structure

3.1 Join the DEC plate to the RA base plate

38. Place the completed RA base plate on the work surface as the bottom of the cube.
39. Engage the DEC plate's rebated locating lip with the matching edge of the RA plate. The geometry only fits correctly in one location. As an orientation reference, when looking down on the RA plate with the RA stepper motor in the top-left corner, the DEC plate attaches to the edge opposite the RA motor.
40. Install three M4 × 16 mm countersunk screws through the DEC plate into the RA plate. At this stage, tighten them only snugly.

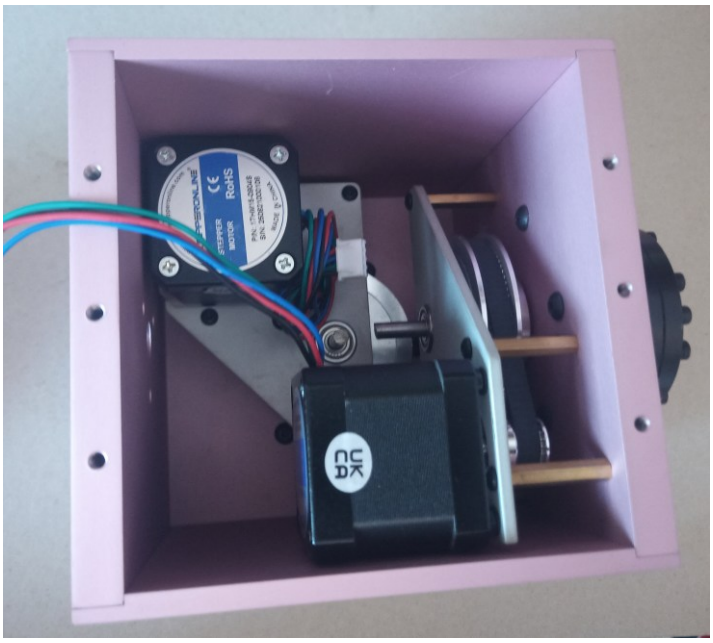


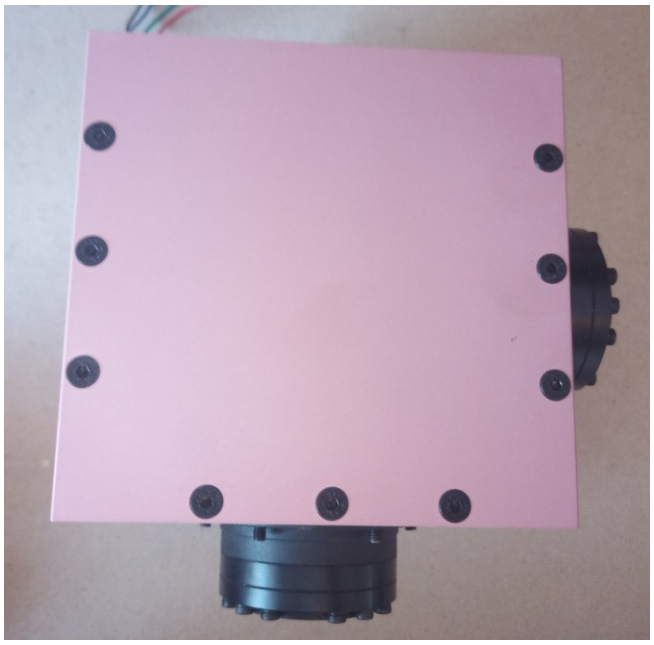
3.2 Fit the rear / counterweight plate

41. Fit the rear/counterweight plate to the RA base plate on the edge opposite the DEC plate. Its rebated lip and mounting pattern permit only the correct orientation.
42. Install three M4 × 16 mm countersunk screws. Leave them snug rather than fully tightened.

3.3 Fit the left and right side plates

43. Fit the left and right side plates into their rebated locating slots. Current STEP files have LEFT and RIGHT engraved into the parts; early prototype plates are unmarked, but the hole patterns only register in the correct orientation.
44. Install nine M4 × 12 mm countersunk screws per side plate (18 total). Keep all side-plate screws snug at this stage so the structure can settle square before final tightening.





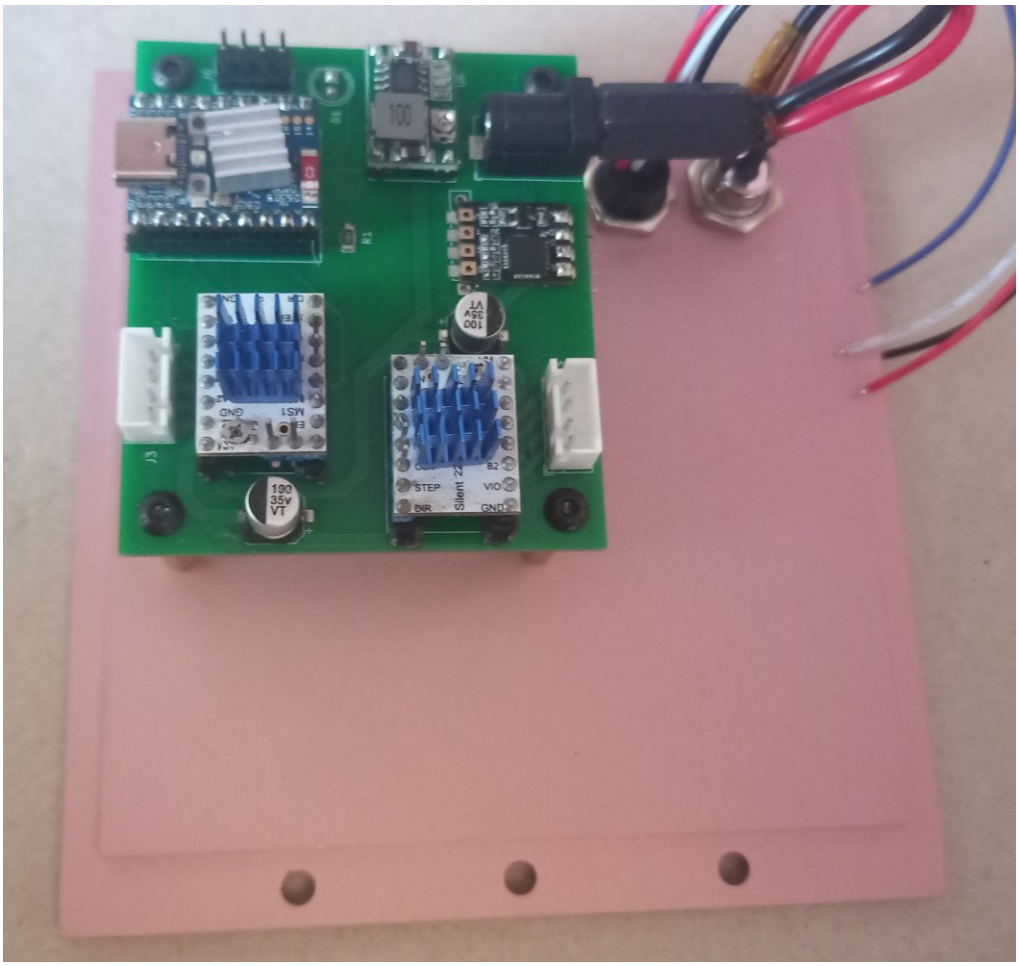
3.4 Square and fully tighten the structure

45. Use an engineer's square or equivalent tool to bring the frame as accurately square as possible. Pay particular attention to the RA-to-DEC plate angle. With accurate machining, the rebated joints should naturally pull into square as the fasteners are tightened.
46. First square the DEC plate to the RA base plate. Fully tighten the three DEC-to-RA M4 × 16 countersunk screws, taking care not to over-torque them.
47. With the DEC/RA angle held square, fully tighten the side-plate screws that attach the side plates to the DEC plate.
48. Next square the rear/counterweight plate. Fully tighten its three M4 × 16 countersunk screws into the RA plate.
49. Fully tighten the side-plate screws that attach the side plates to the rear/counterweight plate.
50. Finally, fully tighten the side-plate screws that attach both side plates to the RA base plate. The main mechanical cube structure is now rigidly assembled.

4. Assemble and install the top plate

4.1 Mount the controller PCB

51. Attach four M3 female–female standoffs to the top plate using four M3 × 8 mm countersunk screws. The demonstrated build uses 24 mm standoffs; any length from 10 mm to 24 mm is acceptable. Avoid exceeding 24 mm because internal space becomes limited.
52. Orient the controller PCB so that its underside faces the top plate and the populated/component side faces down into the cube. The power and USB headers should face into the available open internal space, and the ESP32-S3 should sit approximately in the corner of the top plate as shown in the reference photograph.
53. Secure the PCB to the four standoffs using four M3 × 8 mm button-head screws.

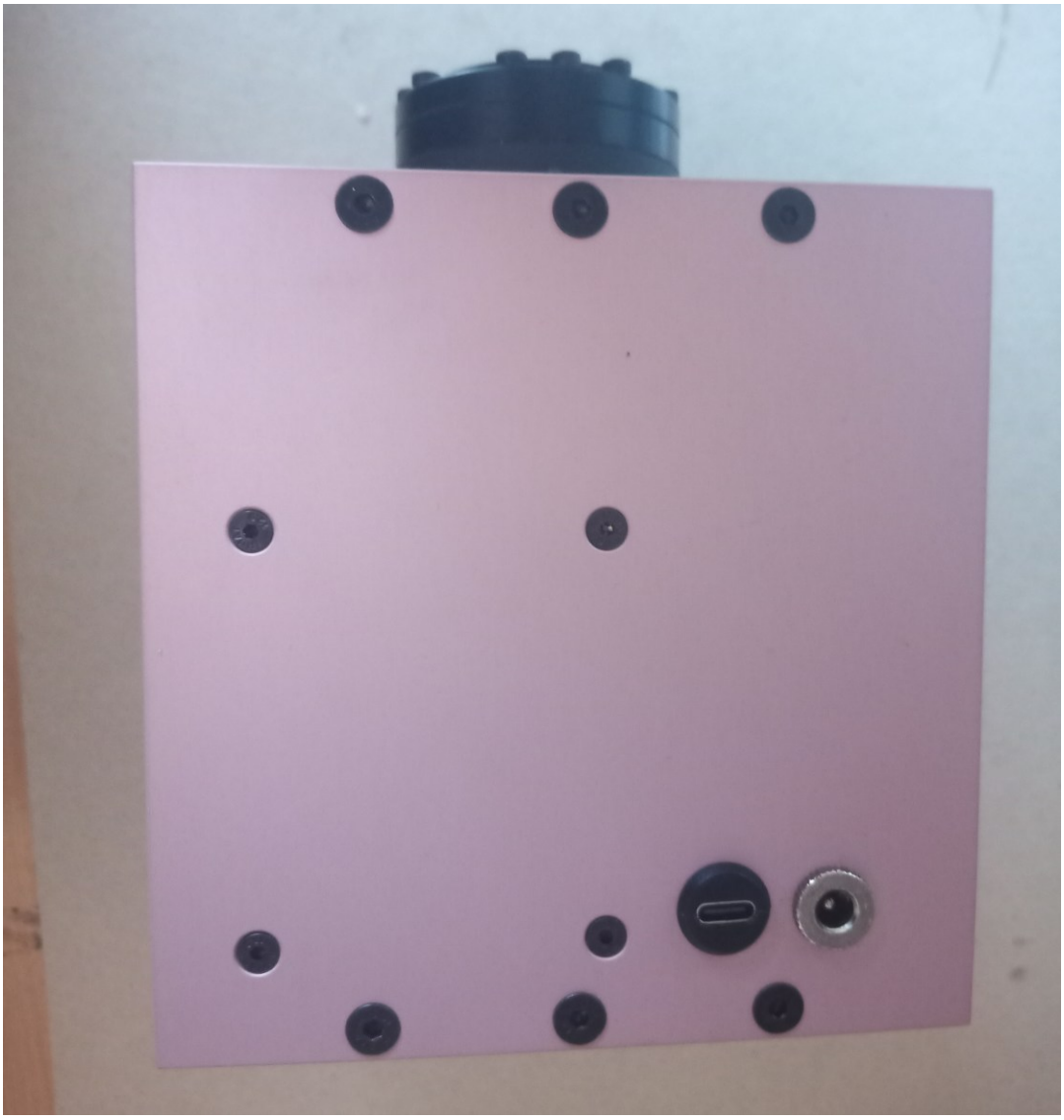


4.2 Fit the power and USB-C panel connectors

54. Fit the chosen panel-mount power connector into its dedicated top-plate opening.
55. Fit the chosen panel-mount USB-C connector into the second dedicated top-plate opening.
56. Wire both connectors to the controller PCB using the connection method of your choice. The demonstrated build uses removable plug-in headers so the top plate can be disconnected easily.

4.3 Connect the motors and close the cube

57. Connect the RA and DEC stepper motors to the controller PCB in accordance with the separate electronics/controller-board instructions.
58. Lower the top plate into place with the panel-mount power and USB-C connectors facing the rear/counterweight side of the CubeMate. This orientation allows the PCB to descend into the available internal space.
59. Seat the top plate fully into its machined rebate.
60. Install six M4 × 12 mm countersunk screws: three into the rear/counterweight plate and three into the DEC plate. Tighten all six fully.



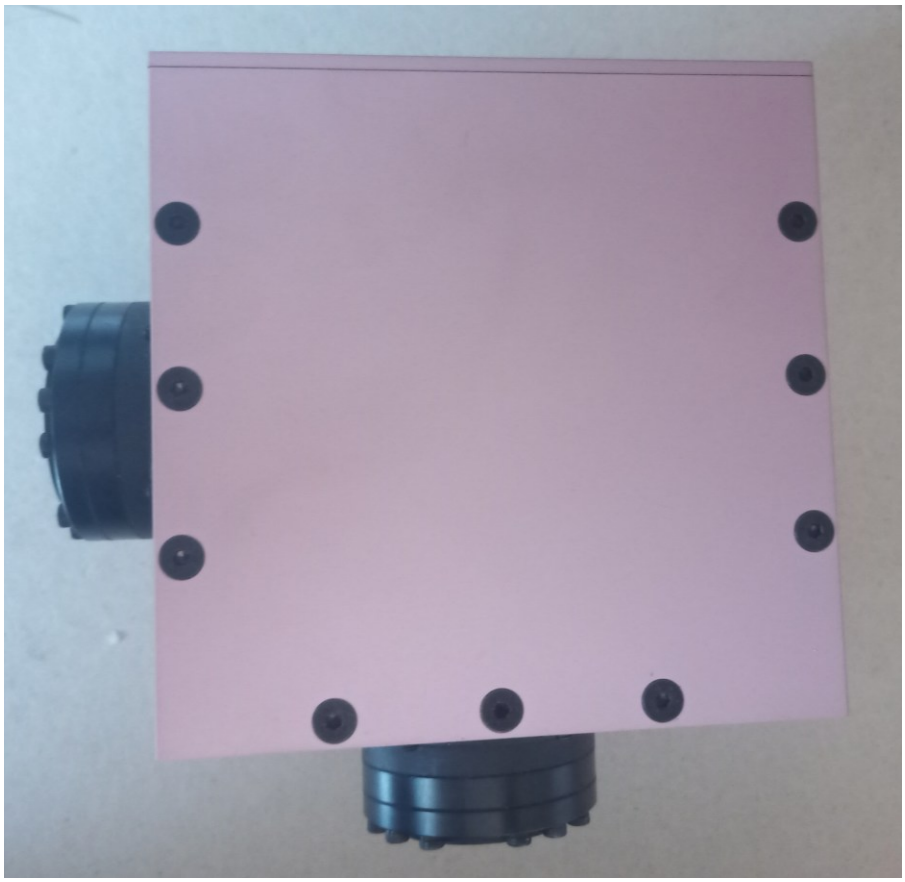
5. Fit the telescope and EQ-wedge interfaces

5.1 DEC telescope dovetail mount

61. Fit the supplied Vixen-style telescope dovetail mount to the DEC axis. A STEP file for this component is included with the project.
62. Secure the dovetail mount using six M4 × 16 mm countersunk screws.

5.2 RA-to-EQ-wedge interface

63. The prototype uses a custom-designed EQ wedge and custom RA interface. That wedge is not currently intended for publication because it is not sufficiently user-friendly.
64. For a published build, a Vixen-dovetail STEP interface is supplied. Alternatively a builder may design an application-specific RA interface.
65. Secure the chosen RA interface plate using six M4 × 16 mm countersunk screws.
66. Attach CubeMate to the selected EQ wedge using the chosen mechanism.



6. Final mechanical checks

- All outer plates are seated fully in their machined rebates/lips and the cube remains square.
- All structural M4 fasteners are fully tightened without over-torquing.
- Both harmonic drives are correctly seated and lubricated.
- Both 80T main pulleys are secured to their 5 mm shafts with their grub/set screws.
- RA and DEC pulley pairs are axially aligned and both GT2 belts have moderate, not excessive, tension.
- Both motor mounting plates are fully secured at all five M3 × 16 points.
- Motor wiring clears both drive assemblies and cannot foul the belts or pulleys.
- Top-plate PCB and panel connectors clear the internal mechanisms.
- DEC payload dovetail and RA wedge-interface plate are securely attached.

Assembly complete: At this point the CubeMate mechanical assembly is complete. Controller configuration, firmware setup, motor-driver settings, commissioning, polar alignment and the separate EQ-wedge build are outside the scope of this mechanical guide.